H

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME	
CLASS	

BIOLOGY **9648/02**

Paper 2 Core Paper **26 August 2016**

Additional Materials: Answer Paper **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.
Circle the question number of the question attempted.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
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Section B	
9 / 10	
Total	

This document consists of **20** printed pages and **2** blank pages.

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Section A

Answer **all** the questions in this section.

1 Cathepsin is a class of protease that is commonly found in the lysosomes of all animals.

Different subtypes of cathepsin had been identified, each with its specific substrate and optimum pH. Fig 1.1 shows the relative activity of three subtypes of cathepsin at different pH.

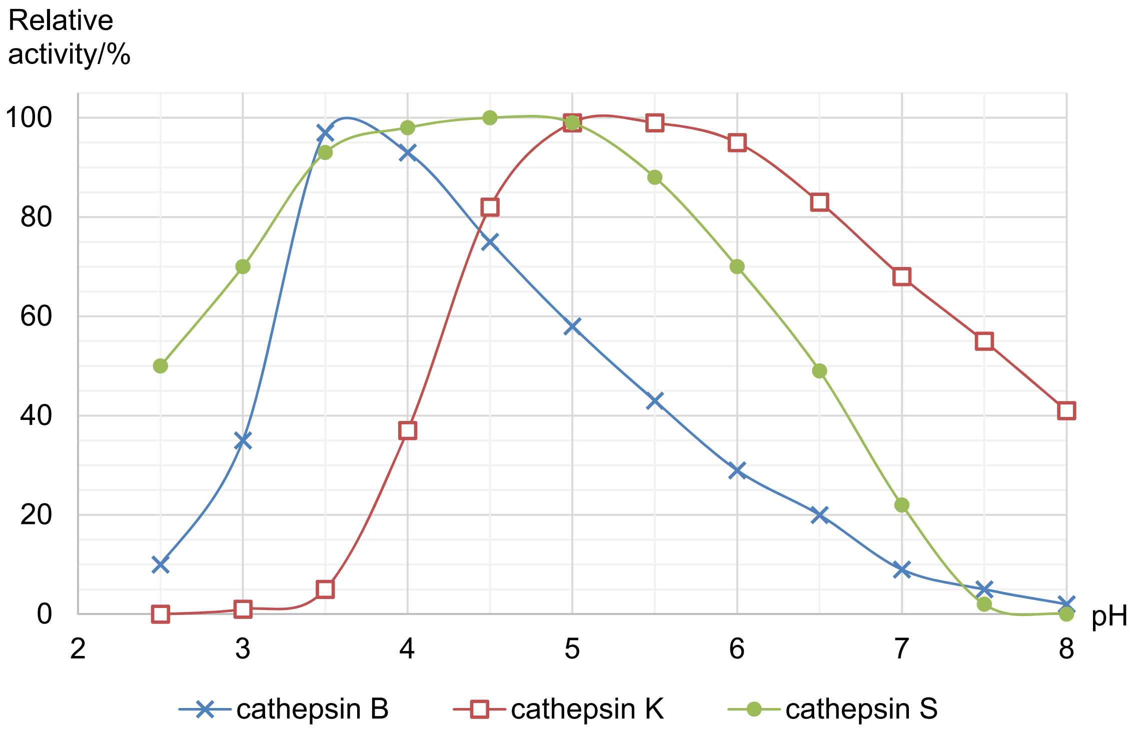

Fig. 1.1

(a) With reference to Fig 1.1,

(i) explain why cathepsin K is not active at pH 3; [3]

(ii) state the optimum pH of both cathepsin B and cathepsin S; [1]

(iii) explain the difference in optimum pH for cathepsin B and S using your knowledge of protein structure. [2]

Different subtypes of cathepsins are expressed in different cell types. There is sufficient evidence that overexpression of cathepsin leads to diseases. For example, high level of cathepsin K has been implicated with development of osteoporosis.

For the purpose of treatment, inhibitor molecules are designed to specifically target cathepsin. These inhibitors are categorised into two broad groups: competitive inhibitors or non-competitive inhibitors.

(b) Fig. 1.2 shows effect of substrate concentration on the activity of cathepsin in the absence of inhibitor.

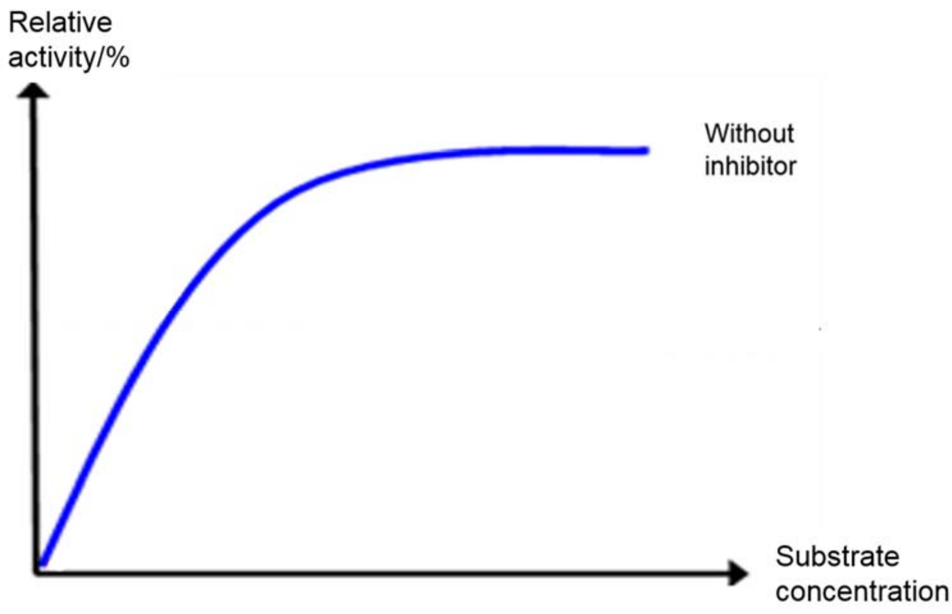

Fig. 1.2

(i) On Fig. 1.2, draw the curve you would expect in the presence of a non-competitive inhibitor. [1]

(ii) Explain with reasons the shape of the curve you have drawn. [3]

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[Total: 10]

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2 Fig. 2.1 shows the structures of the amino acids, glycine and alanine.

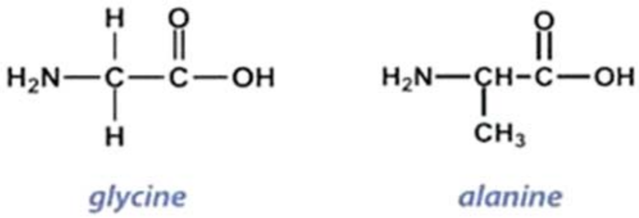

Fig. 2.1

(a) Draw in the space below, the reaction that forms a dipeptide, glycylalanine, containing the above molecules. Label the reactants and products of the reaction clearly. [2]

Glycine and alanine are two of the many amino acids in the structure of haemoglobin, a red pigment that functions in the transport of oxygen in red blood cells. Haemoglobin is a soluble and stable molecule in solution.

(b) Explain how the properties and the location of the amino acid residues in a haemoglobin molecule help maintain the solubility and stability of haemoglobin in solution. [3]

(c) Haemoglobin consists of 2 α-globin chains (each of 141 amino acids) and 2 β-globin chains (each of 146 amino acids). The *HBA* gene that codes for the α-globin protein is found on chromosome 16 of humans and the *HBB* gene that codes for the β-globin protein is found on chromosome 11. The mRNA molecules involved in the synthesis of the globin proteins are shorter compared to the original *HBA* and *HBB* genes.

(i) Explain why the mRNA molecules are shorter compared to the original *HBA* and *HBB* genes. [1]

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(ii) State other changes that can occur to mRNA molecules after they are synthesised. [1]

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In some forms of β-thalassaemia, an inherited blood disorder characterised by the abnormal formation of haemoglobin, the translation of the β-globin gene is affected. Truncated forms of the β-globin polypeptides were found in thalassaemia patients.

(d) Using your biological knowledge,

(i) describe the termination stage during normal translation of the β-globin mRNA; [2]

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(ii) suggest and explain the type of mutation that caused the presence of the truncated forms of β-globin polypeptides found in the thalassaemia patients. [1]

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[Total: 10]

3 Fig. 3.1 shows the main structural features of a mature (infective) human immunodeficiency virus (HIV).

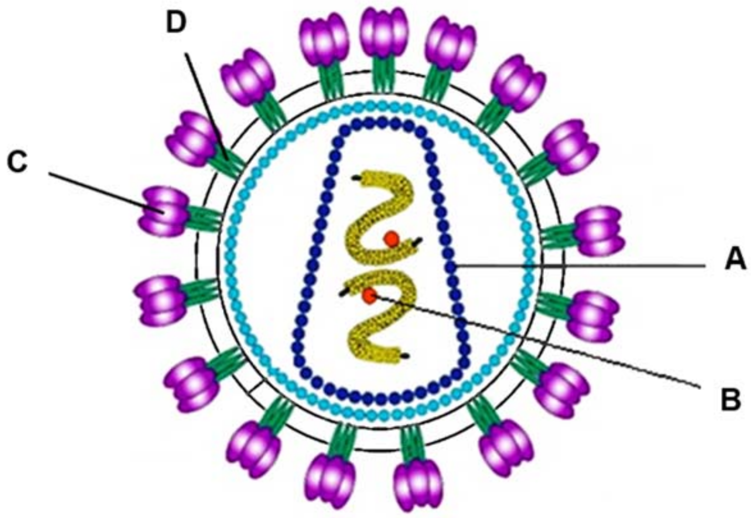

Fig. 3.1

(a) Identify the structure **A** and **B**. [2]

A ____

B ____

(b) Describe the role of **C** and **D** during the entry of HIV into T cells. [3]

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(c) As the global incidence of HIV exceeds 2 million new infections annually, effective interventions to decrease HIV transmission are needed. Truvada is a combination tablet contains two different types of drugs: tenofovir and emtricitabine. Both drugs inhibit reverse transcriptase using different mechanisms.

Studies have demonstrated that daily consumption of Truvada tablets protects healthy individuals against HIV infection by reducing the number of virus copies in the blood.

(i) Using the information above and your knowledge of HIV life cycle, explain how Truvada reduces the number of virus copies in the blood. [2]

(ii) Suggest why two inhibitors with different mechanisms are combined into Truvada. [2]

Fig. 3.2 shows the structure of a Zika virus (ZikV). Most ZikV infection shows no symptom while other infections result in Zika fever which is characterised by mild symptoms such as fever, conjunctivitis (eye infection) and joint pain.

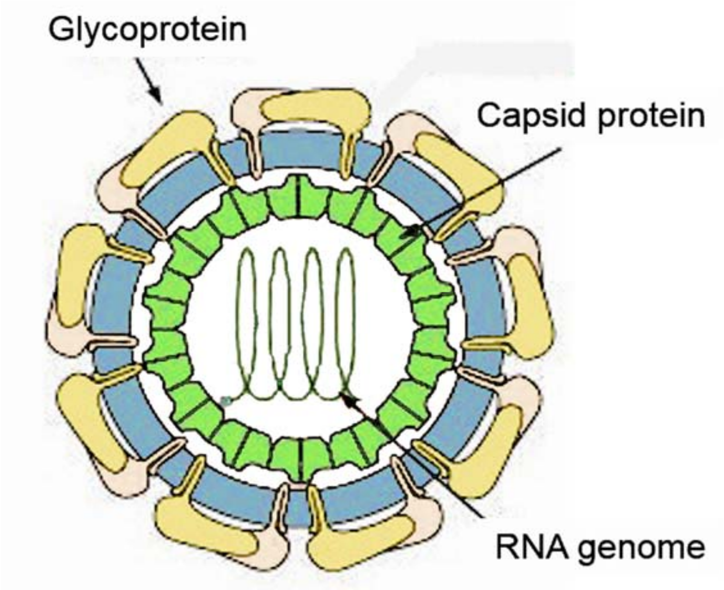

Fig. 3.2

(d) With reference to Fig. 3.1 and Fig. 3.2, state two structural differences between HIV and ZikV. [2]

(e) During the ongoing ZikV outbreak, the incidence of Guillain-Barré syndrome had increased by four times. Guillain-Barré syndrome is a disorder in which the immune system attacks peripheral nervous system leading to progressive muscle weakness.

Using the information above, suggest an explanation on how ZikV infection could have resulted in higher incidence of Guillain-Barré syndrome. [1]

[Total: 12]

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4 The Philadelphia chromosome is an abnormality in chromosome 22 found in chronic myeloid leukemia. This chromosome is defective and unusually short because of reciprocal translocation between chromosome 9 and chromosome 22.

Fig 4.1 shows the chromosome translocation.

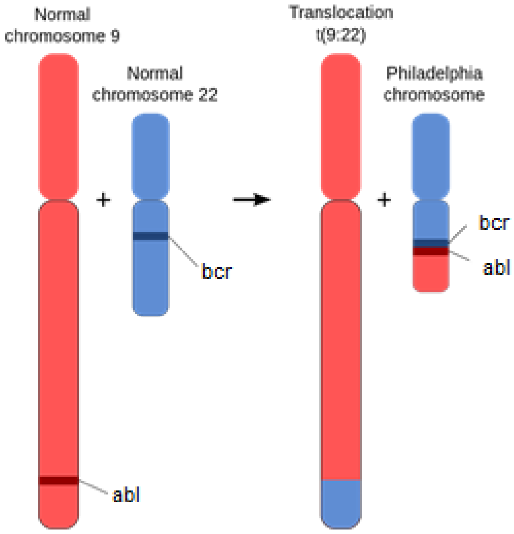

Fig. 4.1

abl is a proto-oncogene found on chromosome 9. Translocation results in an oncogene, *bcr-abl* gene fusion that can be found on the shorter derivative chromosome 22.

Majority of patients with chronic myeloid leukemia have breakpoints in part of *abl* gene and in part of *bcr* gene resulting in *bcr-abl* gene fusion. The bcr-abl transcript is translated into a tyrosine kinase that is constitutively active.

(a) State what is meant by an oncogene. [2]

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(b) Explain how the *abl* proto-oncogene becomes an oncogene in chronic myeloid leukemia cells. [3]

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The *H-ras* oncogene codes for the ras protein.

Fig.4.2 is part of the base sequence of a proto-oncogene and of the *H-ras* oncogene, which arises from the proto-oncogene by mutation.

Fig. 4.2 also shows part of the primary structure of their two encoded proteins.

proto-oncogene

ATG ACG GAA TAT AAG CTG GTG GTG GTG GGC GCC GGC GGT GTG GGC

met thr glu tyr lys leu val val val gly ala gly gly val gly

encoded protein

oncogene

ATG ACG GAA TAT AAG CTG GTG GTG GTG GGC GCC GTC GGT GTG GGC

met thr glu tyr lys leu val val val gly ala val gly val gly

encoded protein

Fig. 4.2

(c) Predict how ras protein differs from the normal protein encoded for by proto-oncogene. [1]

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(d) With reference to Fig. 4.1 and Fig. 4.2, describe how this mutation of proto-oncogene to oncogene differs from the mutation that involves the *bcr-abl* oncogene. [2]

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[Total: 8]

5 Onion bulbs may be red, white or yellow. These colour differences are under the control of two pairs of genes.

A farmer crossed a plant with white onions and a plant with red onions. All F_1 offspring were plants with red onions. The offspring were selfed and the results of the cross were:

Plants with red onions	182
Plants with yellow onions	65
Plants with white onions	82

(a) Using a genetic diagram, illustrate the selfing of the F_1 generation to give the F_2 generation. [4]

Use the following symbols:
A represents allele coding for enzyme which deposits pigment
a represents allele coding for non-functional enzyme for pigment deposition
B represents allele for red pigment
b represents allele for yellow pigment

(b) A chi-squared test was performed on the results of the cross.

(i) Calculate the χ^2 value by filling in the Table 5.1. [2]

Table 5.1

	Observed (O)	Expected (E)	$\frac{(O-E)^2}{E}$	$\sum \frac{(O-E)^2}{E}$
Plants with red onions	182			χ^2 value =
Plants with yellow onions	65			
Plants with white onions	82			

(ii) Suggest reasons why the actual numbers observed do not match the expected ratio exactly. [2]

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(iii) Using Table 5.2, explain if the observed numbers conformed to the expected. [2]

Table 5.2

degrees of freedom	probability P				
	0.50	0.10	0.05	0.01	0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.35	16.27
4	3.36	7.78	9.49	13.28	18.47

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[Total: 10]

6 Telomeres are DNA nucleotide sequences found at both ends of chromosome. The telomeres in somatic cells will get slightly shorter with each round of cell division until a critical short length is reached. Critically short telomeres trigger activation of p53 protein which is required for the expression of a pro-apoptosis gene, P53 Up-regulated Modulator of Apoptosis (*puma*).

Fig. 6.1 shows part of the process, transcription of *puma* taking place.

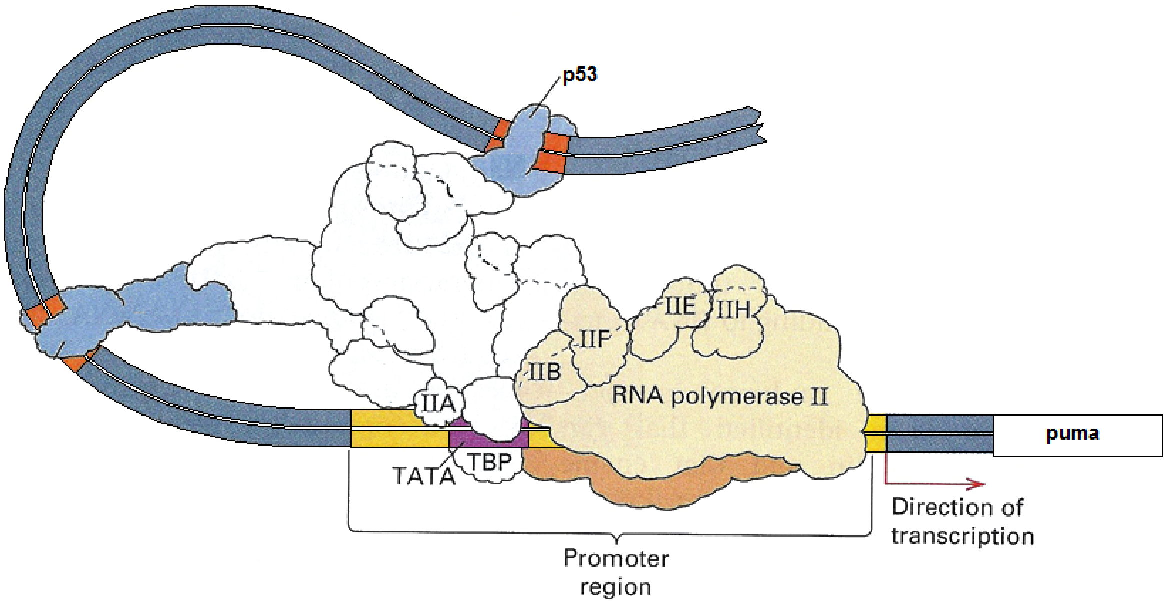

Fig. 6.1

(a) With reference to Fig. 6.1,

(i) suggest the role of p53; [1]

(ii) describe the initiation of transcription of *puma*. [4]

(b) Modifications of DNA or histones facilitate the expression of *puma*.

Outline how two modifications of DNA or histones can lead to expression of *puma*. [3]

(c) Explain the significance of activating p53 in the presence of critically short telomeres. [2]

[Total: 10]

7 Fig. 7.1 shows part of a wheat leaf cell under high power magnification of an electron microscope.

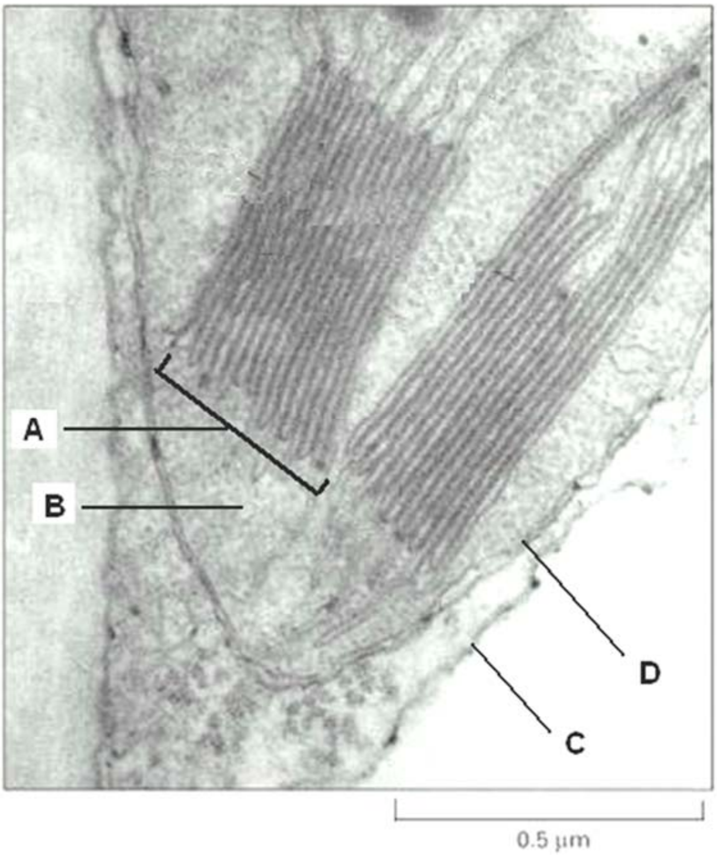

Fig. 7.1

(a) Identify the structures labelled **A** to **D**. [2]

A ______________________________

B ______________________________

C ______________________________

D ______________________________

In wheat plants, the light-dependent reactions are carried out by photosynthetic pigments which fall into two categories: primary pigments and accessory pigments.

(b) Using your biological knowledge,

(i) explain the significance of photosynthetic membranes in chloroplasts to photosynthetic pigments; [1]

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(ii) outline the role played by accessory pigments in the light-dependent reactions. [2]

(c) Wheat is usually grown in the open fields for agricultural purposes. During one of the growing seasons, the skies were unusually dark and gloomy. It was observed that the wheat was not able to develop fully.

Suggest a possible explanation for the above observation. [3]

NADP is an important molecule in the light-dependent and light-independent reactions of photosynthesis.

(d) Outline the role of NADP in wheat leaf cells. [2]

[Total: 10]

8 Anole lizards are found throughout the Carribbean and the surrounding mainland. There are many species of these lizards. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida, USA.

Fig. 8.1 shows the distribution of four species of anole lizards.

Fig. 8.1

An investigation was carried out into the relationships between these four lizard species, using DNA analysis. The base sequences of a region of mitochondrial DNA (mtDNA) from the four species were compared. The results are shown in Table 8.1.

The smaller the number in the boxes, the smaller the percentage differences between the base sequences of the two species compared.

	A. brunneus			
A. brunneus		***A. smaragdinus***		
A. smaragdinus	12.1		***A. carolinensis***	
A. carolinensis	16.7	15.0		***A. porcatus***
A. porcatus	11.3	8.9	13.2	

Table 8.1

(a) With reference to Table 8.1, state the species to which *A. brunneus* appears to be most closely related. [1]

(b) Researchers put forward the hypothesis that the three species, *A. brunneus, A smaragdinus and A. carolinensis*, have originated from three **separate** events in which a few individuals of *A. porcatus* spread directly from Cuba to three different places.

Explain how the results in Table 8.1 support the researchers' hypothesis. [3]

(c) Explain how a population of *A. porcatus* that became isolated on an island could have evolved into a new species. [4]

(d) Explain why some phylogenetic studies use mitochondrial DNA (mtDNA) from mitochondria rather than nuclear DNA. [2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Describe how the chromosomes behave during mitotic cell cycle. [6]

(b) Outline the role of mitosis in multicellular organisms. [8]

(c) Describe the structure and function of centromere. [6]

[Total: 20]

10 **(a)** Explain the principles of homeostasis. [8]

(b) Explain the role of glucagon in regulating blood glucose concentration in humans. [6]

(c) Explain how a nerve impulse is passed across a synapse. [6]

[Total: 20]